# EFCOMFF-Net: A Multiscale Feature Fusion Architecture With Enhanced Feature Correlation for Remote Sensing Image Scene Classification

Junsong Chen, Jizheng Yi, *Member, IEEE*, Aibin Chen, and Ze Jin

***Abstract*— Remote sensing images have the essential attribute of large-scale spatial variation and complex scene information, as well as the high similarity between various classes and the significant differences within the same class, which are easy to cause misclassification. To solve this problem, an efficient systematic architecture named multiscale feature fusion network with enhanced feature correlation (EFCOMFF-Net) is proposed to reduce the gap among multiscale features and fuse them to improve the representation ability of remote sensing images. First, to strengthen the correlation of multiscale features, a feature correlation enhancement module (FCEM) is specifically developed, which takes the features of different stages of the backbone network as input data to obtain multiscale features with enhanced correlation. Considering the differences between the shallow features and the deep features, the EFCOMFF-Net-v1 related to shallow features and EFCOMFF-Net-v2 related to deep features with different structures are proposed. Second, the designed two versions of the deep learning network focus on the global contour information and need to encode more accurate spatial information. A feature aggregation attention module (FAAM) is designed and embedded into the network to encode the deep features by applying the spatial information aggregation features. Finally, considering that the simple integration strategy cannot reduce the gap between the shallow multiscale features and the deep features, a feature refinement module (FRM) is presented to optimize the network. ResNet50, DenseNet121, and ResNet152 are selected to conduct a considerable number of experiments on four datasets, which show the superiority of our method compared to recent methods.**

***Index Terms*— Attention mechanism, feature fusion and correlation, multiscale feature extraction, remote sensing scene classification.**

## I. Introduction

EXTRACTING valuable information from high-resolution remote sensing images and automatically classifying them of different scenes are hot topics of recent interest. Since remotely sensed images are captured in the air using satellites, they are characterized by rich spatial detailed information and diverse feature and thus have been applied in many fields, such as image caption [1], road extraction [2], and target detection [3]. However, the complexity of remote sensing image scenes poses a challenge to the classification task.

### A. Intraclass Variability and Interclass Similarity

As the characteristics of remote sensing images include enormous spatial scope, complex scenes, and the great detail between different features, even the same geospatial object has different shooting angles and sizes, thus resulting in a serious phenomenon of "same object different spectrum, different object same spectrum." In the face of complex and clear high-resolution remote sensing images, it is difficult to extract matched semantic information of various objects effectively.

### B. Insufficient Extraction of Local Semantic Information

In recent years, the advantages of convolutional neural network (CNN) in image processing have been widely explored, but the limitations of CNN model are still apparent. Currently, those CNNs which focus on deep semantic information and consider the key local information related to scene classification are proven to be able to obtain more distinguishing features [4].

### C. More Discriminative Features Are Needed

Recent studies have shown that more discriminative features improve the network's recognition accuracy for complex scenes [5]. In addition, to overcome the problem of narrow field of view for information collection due to the limitation of fixed input of CNN model, the use of constructing multiscale CNN feature representation can solve this drawback [6]. As we all know, feature pyramid network (FPN) [6] aims to obtain features with both resolution and semantic information through feature fusion of different scales. However, there are two shortcomings in the construction of FPN. First, there is information redundancy and insufficient integration. FPN fuses the upsampling of deep features with shallow features by addition, which will result in insufficient fusion, and the redundant background information contained in deep features will interfere with the accurate positioning of the target. Second, the FPN is simply formed according to the structure of the backbone network. In actual remote sensing images,

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the size of the target may be different, and the local features extracted from the shallow layer are not enough to describe the target features. Appropriately increasing the receptive field can obtain more detailed information and improve recognition performance.

In view of the above problems, this article fuses multiscale local information and deep global information to improve the recognition accuracy of current CNN models, and a multiscale feature fusion network with enhanced feature correlation (EFCOMFF-Net) is proposed to explore the impact of enhancing multiscale feature correlation to improve the discriminative property of the network on complex remote sensing images.

For different scene elements in remote sensing images, we intercept outputs of different convolutional layers and enhance the diversity of information in each layer by establishing the correlation between shallow and deep features. Considering that the deep layer of the CNN mainly extracts global contour features and needs to encode more accurate spatial information, a feature aggregation attention module (FAAM) is proposed to encode the outputs from deep layers by using a spatial information convergence strategy. Moreover, a feature map with stronger representation capability is drawn by fusing the output logic of different branches. Considering the variability of a spatial distribution to the fused feature maps, a feature refinement module (FRM) is developed to further reduce their spatial gap.

To investigate the effects of shallow feature correlation and deep feature correlation on the classification performance, two models named EFCOMFF-Net-v1 and EFCOMFF-Net-v2 are designed. The EFCOMFF-Net-v1 model performs feature interaction between "Layer1" and "Layer2" in the feature correlation enhancement module (FCEM), while the EFCOMFF-Net-v2 model performs information interaction between two or three layers of "Layer1," "Layer2," and "Layer3." Of course, this kind of model construction method will increase the parameter quantity due to the changes of scale and channel. To investigate the feature extraction capability of these two models for different benchmark networks, we adopt ResNet50 [7], ResNet152 [7], and DenseNet121 [8] as backbone networks on aerial image dataset (AID) [9], UC-Merced (UCM) [10], OPTIMAL-31 [11], and NWPU-RESISC45 (REmote Sensing Image Scene Classification (RESISC), created by Northwestern Polytechnical University (NWPU)) [12] datasets to test the overall accuracy (OA) of both approaches. The experimental results prove that strengthening shallow information correlation is better than strengthening deep information correlation for classification. To show the advancement of the proposed models, we compare the proposed EFCOMFF-Net with the current state-of-the-art methods and achieve optimal performance in several datasets. The main contributions are summarized as follows.

1) To explore the impact of enhancing multiscale feature correlation on improving classification performance, the FCEM is proposed to achieve information interaction, narrow the gap between multiscale features, and exhibit excellent feature extraction capabilities for being embedded into some existing backbone networks.
2) For accurately encoding the spatial information of deep features, the FAAM is developed to weigh outputs of the final layer of the backbone network for further converging spatial features.
3) Because the multiscale characteristics of fused FCEM and FAAM outputs are different in spatial distribution, the FRM is designed to recode the spatial information and reduce variability among the features through smooth transition.

The rest of this article is organized as follows. Section II reviews some works related to this study. Section III describes the design process of our proposed EFCOMFF-Net. Section IV presents the experimental procedure, results, and visualization effects. Section V concludes and hints at the direction of future work.

## II. Related Work

### A. Remote Sensing Image Scene Classification

Feature extraction algorithms are critical to the success or failure of remote sensing image scene classification, and methods in this field can be roughly divided into three categories including low-level feature extraction, middle-level feature extraction, and high-level feature extraction.

In the early image scene classification methods, low-level features are widely adopted, which are generally designed by researchers according to image contents, so they are usually intuitive and simple. The manual low-level features from the optical remote sensing images are classified into three categories including spectral parameters, texture distribution, and structural characteristics. The spectral parameter is based on the pixel brightness of remote sensing image, and texture distribution and structural characteristics are designed based on the distribution of spatial information (also known as spatial features) which are usually sensitive for the image with prominent visual structure. As the spectral parameter, gray values [13], average values [14], and variance values [15] are usually calculated as the input features for the classification of remote sensing image scenes. For spatial features, the Markov random field (MRF) model [16], gray level co-occurrence matrix (GLCM) [17], and wavelet transform (WT) [18] are usually adopted to extract spatial information for classification. Although manual extraction of low-level features is simple, most hand-made ones are generally designed for specific tasks and depend on expert knowledge in the parameter setting stage. The description capability of hand-made features is not enough to distinguish subtle changes between different categories or massive differences between the same categories, which limits the applicability of these methods under challenging scenarios.

The method based on middle-level features is to bridge the so-called semantic gap between low-level features and semantic tags, and encodes the local descriptors and transforms them into global features that describe the whole image. Common feature coding methods include bag of visual words (BoVW) [19], principal component analysis (PCA) [20], and potential latent semantic analysis (PLSA) [21]. These methods are superior to those based on low-level features in classification performance but have the faultiness of complicated computation and poor adaptability.

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Therefore, high-level feature (rich semantic information) extraction based on deep learning is widely adopted to capture the most essential distinguishing information of remote sensing images and has made remarkable achievements. According to the number of training samples, deep learning methods are divided into fully supervised learning, semisupervised learning, and unsupervised learning.

Fully supervised learning means that each image corresponds to a unique label, and needs to train a large amount of data with labels. Liu et al. [22] proposed a deep random-scale stretched CNN (SRSCNN) to improve the robustness of the multiscale features, thus reducing the classification error rate. Huang et al. [23] proposed a critical feature sparse layer, which suppresses confusing features and emphasizes distinguishing features, and used it to extract multilayer feature maps from sparse convolution layers. Since it takes a lot of time and training cost to completely design and train a CNN model for remote sensing scene classification. For improving the training speed and classification accuracy, many methods are fine-tuned based on the traditional CNN models. Muhammad et al. [24] adopted pretrained visual geometry group (VGGNet) as a depth feature extractor to capture intermediate and depth features of remote sensing scene images. Bei et al. [25] fine-tuned the fully pretrained convolutional networks (CNNet) by random clipping and mirroring strategies to speed up the convergence of the training process. Scott et al. [26] classified the remote sensing images by optimizing their CNN architecture. The fully supervised classification method based on deep learning shows superior performance, but its disadvantage is that the training is based on plenty of labeled samples. In reality, the number of unlabeled images is far more than that of labeled images.

For improving the utilization rate of the unlabeled samples, the semisupervised learning methods combined with both labeled and unlabeled samples have attracted more and more attention. Miao et al. [27] proposed a semisupervised twin network combined with the mixed-labeled and pseudo-labeled sample training networks. Lasloum et al. [28] presented a method of multisource semisupervised domain adaptation, which adopts adversarial learning to train models and combines cross-entropy loss and minimax loss for image classification. Zheng et al. [29] developed a preprocessing solution based on semisupervised learning called NearPseudo and created a feedback mechanism based on consistency check to select unlabeled data with pseudo-labels effectively. Their work solved the problem of sample imbalance caused by a few sparse samples and improved the classification accuracy.

It is easy to obtain remote sensing images, but it is difficult to label them. Recent years have witnessed that a large number of unsupervised learning methods shine brilliantly in the field of remote sensing image classification. Typical unsupervised feature learning methods include K-means clustering [30], generative adversarial network (GAN) [31], and dimension reduction [32]. Zhang et al. [33] improved the existing data clustering algorithm, proposed a unified low-rank matrix decomposition, and combined dimensionality reduction and data clustering, which was superior to the existing clustering methods. Li et al. [34] designed a variational GAN that adopts the proposed two-branch variational encoder to map spectral and spatial information to a specific space, combined with a cross-interactive generator to improve image quality and achieve classification by ensuring diversity as well as interacting spectral and spatial features.

### B. Multiscale Feature Fusion

It is generally believed that the strategies of multiscale feature fusion for different types of images including remote sensing images should be universal. As another critical research content in this work, the studies about multiscale feature fusion are reviewed in the following paragraphs.

The fusion of features with different scales is beneficial to organically combine the advantages or complementarities of each image, enhance the diversity of features, and improve the generalization ability of the model for remote sensing images with complex scenes. Recently, in the task of remote sensing image classification, many researchers have improved the classification performance of models by fusing multiscale features [35], [36], [37], [38], [39], [40].

Some works adopt attention mechanisms to weigh multilevel features to improve the representation ability of the learning model. Bai et al. [35] extracted the spatial information from multilevel features by efficient spatial pyramid attention (ESPA) and the fine-grained features from multiscale depthwise network (MSDWNet). Lv et al. [36] designed an aggregation learning strategy to aggregate spatial features and proposed a context-gating mechanism to fuse the aggregated features of local information in different scales. In addition, Lei et al. [37] built a multimodal and multiscale attention fusion network. In their work, on the one hand, a multimodal fusion module was designed to fuse more valuable features. On the other hand, a multiscale spatial context enhancement module was proposed to improve feature fusion, which solved the defect of multiscale change of objects in remote sensing images. Three weighting coefficients are introduced in the work of Xu et al. [38] to integrate multiple characteristics (i.e., multilayer convolution features and fully connected features), and a linear classifier is then used to predict the query image.

Other works focus on processing channels to get more detailed information. Wang et al. [39] designed adaptive channel dimension reduction methods to enhance channel attention and add semantic information, and then reduce the multistage and adaptive channel characteristics together. Zhao et al. [40] proposed a multigranularity multilevel feature ensemble network (MGML-FEN), which obtains structural information and rich fine-grained features through a full channel feature generator (FC-FG) to realize diversified prediction.

Different from the above reviewed methods, the proposed EFCOMFF-Net focuses on the multiscale feature correlation to enhance the generalization of complex scenes and adopts an effective fusion strategy for the shallow and deep features to enhance the representation ability of the network.

## III. Methodology

### A. Overall Architecture

In this study, two kinds of EFCOMFF-Net are designed for increasing the difference between different categories of

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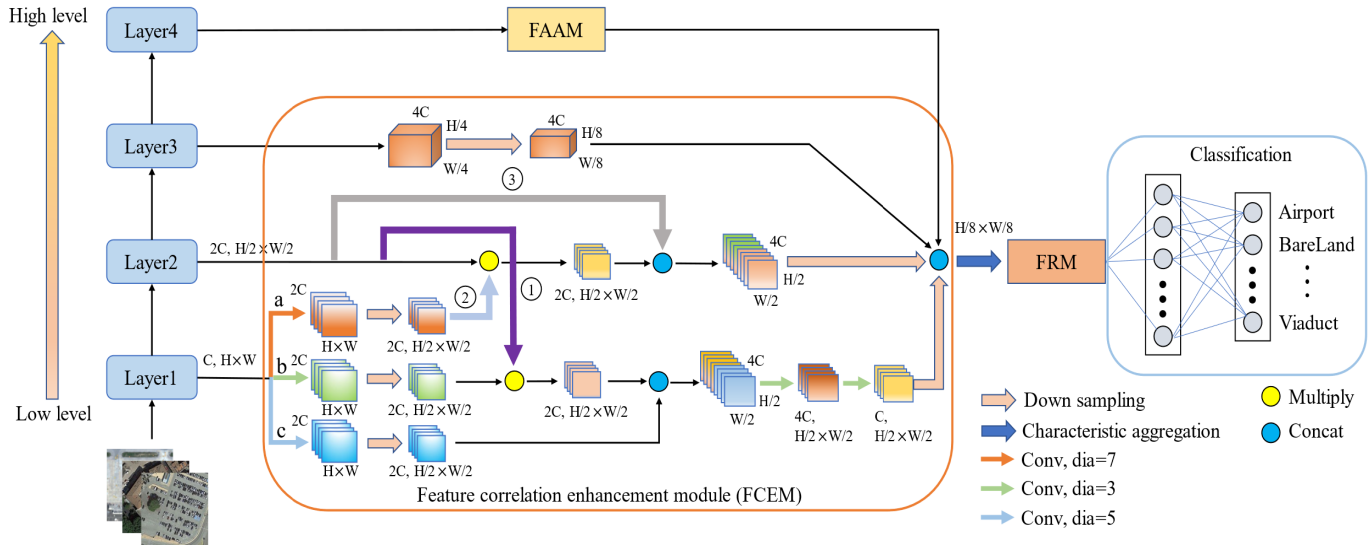


Fig. 1. Overall architecture of EFCOMFF-Net-v1. FCEM is used to narrow the gap between the adjacent multiscale features and make them diversified and highly correlated. FAAM enhances the significance of deep semantic information, and the features extracted by FAAM and FCEM are fused and then input into FRM for reintegration and filtering.

remote sensing images and studying the effect of enhancing the correlation of multiscale features on improving classification accuracy. Fig. 1 shows the overall structure of EFCOMFF-Net-v1, "Layer1"–"Layer4" means from the low level to the deep level of the network, and low-level features contain detailed information that is beneficial to improve the classification performance. With the increase of network depth, the area of concern gradually shifts from local texture to deep semantic information [6].

Simply fusing multiscale information will reduce the expressive ability of the network to complex scenes due to the difference in feature space [41], [42]. To solve this problem, FCEM is designed to extract multiscale features. First, to obtain as much image semantic information as possible, the convolution of different dilation rates is adopted to extract the multiview range (different reception fields) output of "Layer1." Second, for reducing the parameter quantity and facilitating the fusion of outputs of different convolution layers, three branches are divided in the proposed FCEM. The downsampling of the branch represented by the "orange feature graph" in Fig. 1 is dot multiplied by the output matrix of "Layer2," and the branch represented by the "green feature graph" is dot multiplied by the output matrix upsampled by "Layer2." After the "blue feature graph" branch being downsampled, it is concatenated with the dot product of the outputs of "Layer1" and "Layer2," which realizes the diversity of feature information. Similarly, the residual structure represented by ③ of "Layer2" in Fig. 1 has the same contribution. The downsampled output of "Layer3" is concatenated with the output of "Layer4" weighted by FAAM. The role of FAAM is to mine the global information of "Layer4," code more accurate spatial information, and further improve the distinguishing ability of salient features. Finally, the outputs of FAAM and FCEM are fused. Because of the spatial difference between shallow features and deep features, the merged results are rough. Hence, the proposed FRM gets a finer feature map and then sends the feature map to the full connection layer for getting the prediction vector. The structures ① –③ in Fig. 1 are represented by three different colors, and the rationality of their design will be discussed in Section IV-F.

Consider that only narrowing the information gap between the first two layers will result in the loss of crucial information in the stage of multiscale information fusion due to the excessive spatial differences. As shown in Fig. 2, EFCOMFF-Net-v2 is updated based on EFCOMFF-Net-v1 to study the impact of strengthening the deep feature correlation on the classification performance. The difference between the two versions is the FCEM structure.

The first improvement is to remove the structure ③ in Fig. 1. The feature map multiplied by the output of "Layer1" and structure ② is first calculated through the dilated convolution with the kernel size of 3 × 3, and then downsampled and multiplied by the output of "Layer3." The second improvement point is that the output of "Layer3" is first performed the dilated convolution with the kernel size of 3 × 3, then upsampled, and multiplied the result which is the product of the output of "Layer1" and structure ②. The remaining structural design of FCEM, FAAM, and FRM is the same as that of EFCOMFF-Net-v1.

Algorithm 1 illustrates the structure and execution of EFCOMFF-Net in pseudo-code, and the individual modules are described in detail in the following.

### *B. Feature Correlation Enhancement Module*

As the network goes deeper, the interesting region of the network gradually goes from local texture to global contour, and the local semantic information contained in the shallow layer is equally important for improving the model that contains accurate positioning of the vital information in the scene. Unlike the previous simple fusion of multiscale features, the proposed FCEM discovers the shallow feature correlation to improve the network representation capability. In Algorithm

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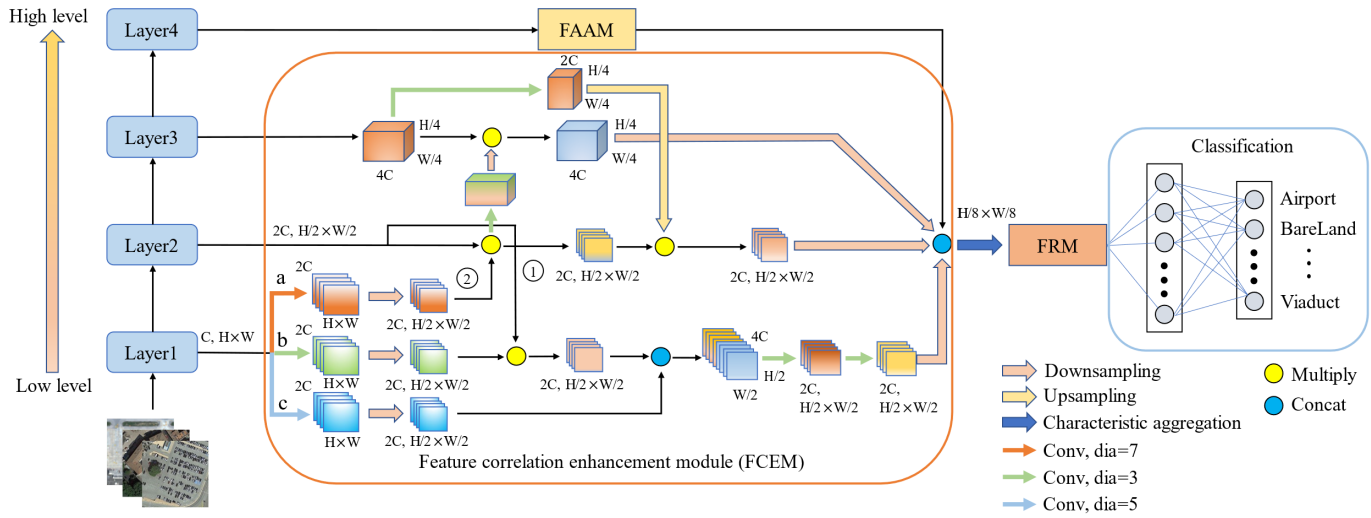


Fig. 2. Overall architecture of EFCOMFF-Net-v2. In order to study the influence of deep feature correlation on classification performance, the proposed FCEM of EFCOMFF-Net-v1 is improved in EFCOMFF-Net-v2. Compared with EFCOMFF-Net-v1, the difference is that the correlations of output features from "Layer1," "Layer2," and "Layer3" are strengthened. The FAAM and the FRM are the same as EFCOMFF-Net-v1 in the structural design.

**Algorithm 1** EFCOMFF-Net

**Input:** A batch of pre-processed images X;
**Output:** Prediction vector $Y$;
**Statement:** $V^m$ represents multi-scale feature extraction, and $V^n$ represents deep feature extraction;
1: **Feature extraction:**
2: $L^s \leftarrow$ Net(X);
3: **Fusion output:**
4: $V^m \leftarrow FCEM(L^S)$;
5: $V^n \leftarrow FAAM(L^S)$;
6: $V^f \leftarrow fusion(V^m, V^n)$;
7: $Y \leftarrow Linear\ (Flatten(Dropout(FRM(V^f))))$
8: **Unit Loss:**
9: $loss \leftarrow CrossEntropyLoss()$;

**Algorithm 2** EFCOMFF-vl-FCEM

**Input:** Feature maps from the branch $L^s$, $s = f_1, f_2, \ldots, f_m$;
**Output:** The output $V^m$ of multi-scale feature fusion;
1: $P_{en} \leftarrow interpolate(Conv(L^{f_1}))$;
2: $O_c \leftarrow interpolate(Conv(Conv((L^{f_2} \otimes P_b) \otimes P_c)))$;
3: $O_t \leftarrow interpolate(L^{f_2} \oplus (L^{f_2} \otimes p_a))$;
4: $O_u\ Batchnorm(avgpool(L^{f_3}))$
5: **Fusion output:**
6: $V^m \leftarrow O_c \oplus O_t \oplus O_u$;

**Algorithm 3** Overall Implementation Process of *Conv*

**Input:** The Arbitrary four-dimensional image $P$;
**Output:** Normalized characteristic diagram $M$;
**Statement:** The "weight" parameter is initialized by Kerning uniform distribution initialization method;
1: $v, m \leftarrow var_mean(weight)$;
2: $W \leftarrow (W - m)/sqrt(v + 1e - 5)$;
3: $P \leftarrow Sc_{3\times3}(P, W)$;
4: $P \leftarrow BatchNorm2\ d(P)$;
5: $P \leftarrow Relu(P)$;
6: $M \leftarrow Dropout(P)$;

2, the FCEM of EFCOMFF-Net-v1 is detailed to describe how to process the multiscale features and calculate the output feature map $V^m$, and each part of the design is explained in the following.

As shown in Fig. 1, the multiscale features of the backbone network are extracted as the inputs of FCEM. The output of the $s$th convolutional layer is defined by the following equation:

$$L^s = \left\{I_1^s, I_2^s, I_3^s, \ldots, I_N^s\right\} \tag{1}$$

where $I_n^s$ is the value of the feature maps at the spatial location, and $n \in \{1, 2, 3, \ldots, N\}$.

From the previous step, the multiscale feature maps from "Layer1" to "Layer4" could be obtained, which are mathematically expressed as $L^{f_1}$, $L^{f_2}$, $L^{f_3}$, and $L^{f_4}$. First, the feature map $L^{f_1}$ is input into FCEM. Since the receptive field of the lower level is small, the proposed dilated convolution is divided into three branches a, b, and c to expand the receptive field. Before merging with the deep features, the number of channels needs to be doubled by convolution, and then, the feature map is compressed by the bilinear interpolation method

$$P_{\text{en}} = \text{Dp}\left(\text{Conv}\left(L^{f_1}\right)\right). \tag{2}$$

Algorithm 3 details the implementation of Conv, where $Sc_{3\times3}$ represents the standard 2-D convolution and is improved by the work of Dosovitskiy et al. [43] based on the traditional 2-D convolution. The kernel size of Conv is $3 \times 3$, the padding and stride are 3, the discard rate is set to 0.5, Dp represents the downsampling, and $P_{\text{en}}$ represents the output feature map in such cases of branch en $= \{a, b, c\}$. When the input $L^{f_1} \in R^{C\times H\times W}$ is passed through (2), the feature map $P_{\text{en}} \in R^{2C\times H/2\times W/2}$ with doubled the number of channels

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**Algorithm 4** EFC0MFF-v2-FCEM

**Input:** Feature maps from the branch $L^s$, $s = f_1, f_2, \ldots, f_m$;
**Output:** The output $V^m$ of multi-scale feature fusion;
1: $P_{en} \leftarrow interpolate\ (Conv(L^{f_1}))$;
2: $O_c \leftarrow interpolate\ (Conv(Conv((L^{f_2} \otimes P_b) \oplus P_c)))$;
3: $O'_t \leftarrow z\ (L^{f_2} \otimes P_a \otimes interpolate\ (Conv(L^{f_3})))$;
4: $O'_u \leftarrow interpolate\ (Conv\ (L^{f_2} \otimes P_a)) \otimes L^{f_3}$;
5: **Fusion output:**
6: $V^m \leftarrow O_c \oplus O'_t \oplus O'_u$;

and size compressions is obtained. To reduce the gap between adjacent features, ① and ② are introduced. The result $O_c$ obtained by fusing the b and c branches and then convolving and downsampling is defined as follows:

$$O_c = \mathrm{Dp}\big(\mathrm{Conv}\big(\mathrm{Conv}\big(\big(L^{f_2} \otimes P_b\big) \oplus P_c\big)\big)\big) \tag{3}$$

where $\oplus$ represents the concatenation function, and $\otimes$ represents the matrix multiplication. The specific operation after the fusion of b and c branches is as follows. After the first Conv convolution operation, the number of channels and the scale of the fused output remain unchanged. The purpose of this design is that the multiple channels could extract features from different regions. After the second Conv, the number of channels is reduced by four times to cut down the number of parameters and then downsample to get the output $O_c \in R^{C\times H/8\times W/8}$. Similarly, the output $O_t \in R^{4C\times H/8\times W/8}$ is obtained by processing the features from "Layer2"

$$O_t = \mathrm{Dp}\big(L^{f_2} \oplus \big(L^{f_2} \otimes P_a\big)\big). \tag{4}$$

For the EFCOMFF-Net-v1, the information sharing among "Layer1," "Layer2," and "Layer3" is not realized, but the output by "Layer3" is downsampled to obtain the feature map $O_u \in R^{4C\times H/8\times W/8}$ which could be fused with $O_c$ and $O_t$. Finally, the output of the feature map $V$ of FCEM in EFCOMFF-Net-v1 which fuses multilevel features in the range $m$ is presented

$$V^m = O_c \oplus O_t \oplus O_u. \tag{5}$$

In addition, for studying the impact of reducing the gap between deep multiscale features on the classification performance, we enhance the connection between "Layer2" and "Layer3" in the FCEM structure based on the EFCOMFF-Net-v1. The implementation of FCEM for EFCOMFF-Net-v2 is shown in Algorithm 4. The improvements are explained in the following.

To interact with the information of "Layer2" and "Layer3," first, the features of "Layer3" need to be transferred to "Layer2" for processing. The specific operation is to reduce the channel multiples by convoluting the output characteristics of "Layer3" with a 3 × 3 convolutional layer. Since the feature size gradually increases from the lower layer to the deeper layer, the feature map after compressing the channel needs to be upsampled again. Hence, the output $O'_t \in R^{2C\times H/2\times W/2}$ is obtained by fusing the features from the two layers. The specific implementation is described as

$$O'_t = \mathrm{Dp}\big(L^{f_2} \otimes P_{a0} \otimes \mathrm{Up}\big(\mathrm{Conv}\big(L^{f_3}\big)\big)\big) \tag{6}$$

where Up represents the upsampling.

Second, we realize the two layers' characteristic information sharing by inputting the feature information of the lower layer "Layer2" into "Layer3." The specific implementation is as follows:

$$O'_u = \mathrm{Dp}\big(\mathrm{Conv}\big(L^{f_2} \otimes P_a\big)\big) \otimes L^{f_3}. \tag{7}$$

Finally, the three layers of outputs with reduced feature gap are fused to obtain another expression of $V^m$ as follows:

$$V^m = O_c \oplus O'_t \oplus O'_u. \tag{8}$$

### C. Feature Aggregation Attention Module

The interesting region becomes a part of the global contour information after the original image passes through the backbone network, which makes the latter crucial to the classification effect. To compress the spatial information to aggregate the features, encode more accurate spatial information, and filter the deep semantic information, a new attention structure named FAAM in this work is presented in Fig. 3.

To reduce the number of parameters in the network, the input feature map is compressed by the 1 × 1 convolution layer, and the channel compression ratio is set to 4. After that, the spatial dimension of the input features is compressed using the average pool and the maximum pool to aggregate the spatial information of the features and improve the representation capability of the network, and finally, two different spatial context descriptors are generated. Unlike the convolutional block attention module (CBAM) [44], a depth-separable convolution with fewer parameters and lower operational cost is introduced in this article to aggregate the features obtained from both pools and generate the attention map for each channel. After normalization, the values of each channel are mapped between 0 and 1 by a Sigmoid function to be weighted to the original input, as implemented in the following equation:

$$V^{t+1} = \delta\big(\mathrm{DSC}\big(F_{\mathrm{Avg}}\big(\mathrm{Con}_{1\times1}\big(V^t\big)\big) \oplus F_{\mathrm{Max}}\big(\mathrm{Con}_{1\times1}\big(V^t\big)\big)\big)\big) \otimes V^t \tag{9}$$

where $\delta$ represents the Sigmoid, $F_{\mathrm{Avg}}$ and $F_{\mathrm{Max}}$ represent the adaptive average pooling and adaptive maximum pooling, respectively, DSC means the depth-separable convolution, $\mathrm{Con}_{1\times1}$ represents the 1 × 1 convolution layer, and $V^{t+1}$ is replaced by $V^n \in R^{c_1\times h_1\times w_1}$.

### D. Feature Refinement Module

Based on the two prediction results $V^m$ and $V^n$ obtained above, the decision fusion of the results is applied. Considering the gap between these two results in spatial features and semantic information, the summation operation would cause some information to be covered, thus losing some crucial information. The splicing operation retains more dimension and location information, which makes the later steps to freely

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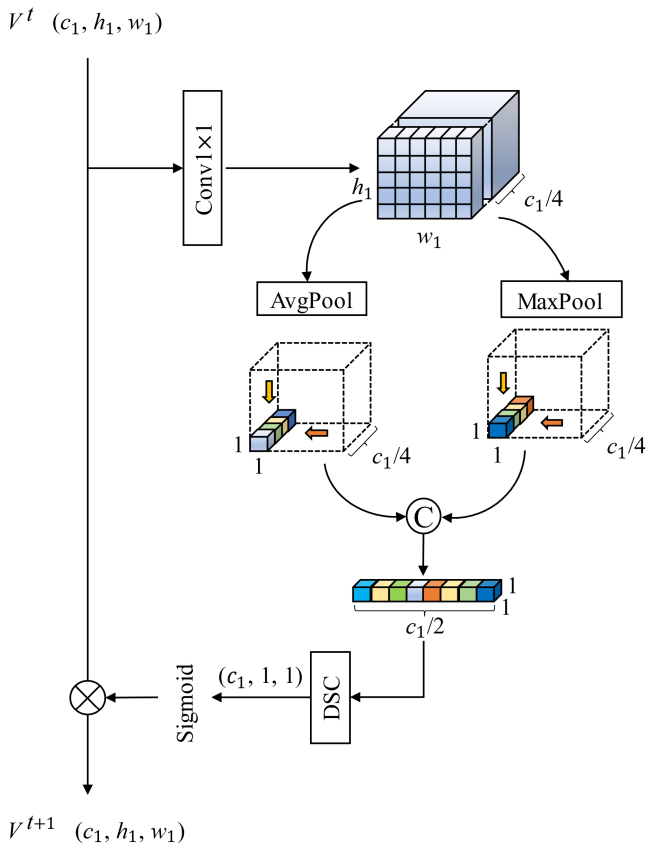


Fig. 3. Overall structure of the proposed FAAM.

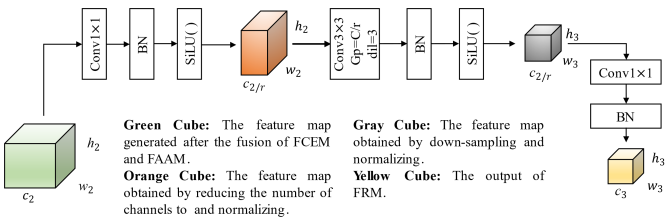


Fig. 4. Overall structure of FRM.

choose between shallow and deep features. Therefore, the concatenation is adopted to fuse these two parts of features

$$V^{\mathrm{f}} = V^m \oplus V^n. \tag{10}$$

However, the fused features are still relatively coarse, and the spatial discrepancy between shallow and deep features needs to be eliminated by a smooth transition process. The FRM is designed as shown in Fig. 4. The output of the previous fusion step is first fed into the $1 \times 1$ convolution layer to compress the channels for reducing the number of parameters. After standardization, the value in the characteristic matrix is limited to $[-0.28, +\infty)$ by the SiLU(). Compared with the ReLU(), the curve is smoother, and more information values are retained if the value is negative. Then, a $3 \times 3$ grouped convolution with a dilation rate of 3 is adopted to obtain a wide range of spatial information while compressing the feature map. In addition, for reducing the number of parameters while improving the classification accuracy, the number of groups is $C/r$. Finally, the feature map is fed into the $1 \times 1$ convolution

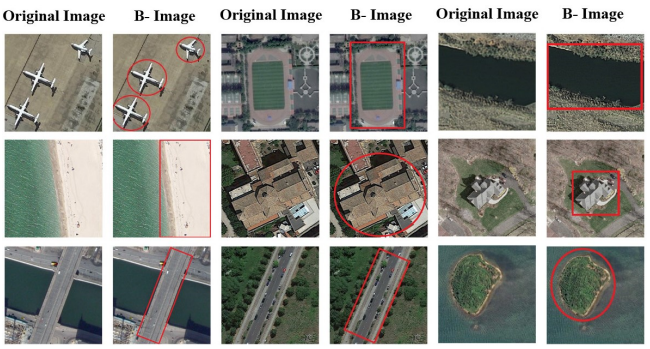


Fig. 5. Some samples of four datasets. B-Image represents the target area of image recognition.

layer to compress the number of channels and then normalized to obtain the feature map $W^l \in R^{1\times h_3 \times w_3}$ at position $l$

$$W^l = \mathrm{Con}_{1\times 1}\left(\varphi\left(\vartheta\left(V_l^{\mathrm{f}}\right)\right)\right) \tag{11}$$

where $\varphi$ represents the traditional convolutional operation with batch normalization and SiLU(), and $\vartheta$ represents the grouped convolutional operation with batch normalization and SiLU(). The feature vector $Q$ is calculated by expanding the feature maps of all $L$ layers

$$Q = \left\{q^1, q^2, q^3, \ldots, q^r\right\} \tag{12}$$

where $r$ represents the number of vectors after expansion. To obtain the prediction result vector, the final prediction is generated by passing the vector $Q$ through the fully connected layer FC

$$Y = \mathrm{FC}(Q). \tag{13}$$

In Section IV, we will adopt a large number of experiments and visual results on four datasets to verify the rationality of EFCOMFF-Net in structure and parameter design.

## IV. Experimental Results and Analysis

### A. Datasets

Fig. 5 shows some examples of four datasets which are defined as follows.

1) The AID [9] dataset consists of 30 remote sensing scenes with at least 200 images in each category. All images are extracted from Google Earth. There are 10 000 images with a resolution of 600 × 600.
2) The UCM [10] dataset consists of 21 remotely sensed urban scenes across the country, with a total of 2100 images extracted from the USGS national map. Each category of remote sensing scene has 100 images with a resolution of 256 × 256.
3) The OPTIMAL-31 [11] dataset consists of 31 remote sensing image scenes; each one contains 60 remote sensing images. These images are extracted from Google Earth, and the resolution of each image is 256 × 256.
4) The NWPU [12] dataset is the largest remote sensing dataset in this work. It contains 45 scene categories. Each one contains 700 images with a resolution of 256 × 256. The spatial resolution ranges from 0.2 to 30 m from air to ground.

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TABLE I
OA OF A DIFFERENT BACKBONE FEATURE EXTRACTION NETWORK ON FOUR DATASETS

| Methods | UCM (OA%) | | AID (OA%) | | NWPU (OA%) | | OPTIMAL (OA%) |
|---|---|---|---|---|---|---|---|
| | Tr=50% | Tr=80% | Tr=20% | Tr=50% | Tr=10% | Tr=20% | Tr=80% |
| **ResNet50** | 97.95 ± 0.12 | 98.93 ± 0.16 | 94.45 ± 0.13 | 96.28 ± 0.12 | 91.24 ± 0.11 | 93.76 ± 0.12 | 92.25 ± 0.17 |
| **ResNet152** | 98.19 ± 0.15 | 98.85 ± 0.12 | 94.56 ± 0.16 | 96.60 ± 0.11 | 91.86 ± 0.17 | 94.20 ± 0.20 | 92.74 ± 0.15 |
| **DenseNet121** | 98.14 ± 0.08 | 98.90 ± 0.20 | 94.21 ± 0.12 | 96.33 ± 0.07 | 90.67 ± 0.13 | 93.49 ± 0.14 | 94.08 ± 0.13 |

Note: OA means Overall Accuracy, Tr means training rate.

### B. Experimental Setup

*1) Experimental Parameter Settings:* We adopt the widely used ResNet50, ResNet152, and DenseNet121 as feature extraction backbone networks, as shown in Table I, to test their classification accuracies on four datasets. In the training, random cropping (stride size set to 4), horizontal flipping, and normalization are adopted to enhance the generalization ability of the model. To adapt to the hardware environmental requirements, the image resolution is adjusted to 224 × 224, the batch size is set to 32, the learning rate is adjusted using a cosine annealing strategy, the initial learning rate is set to 0.0002, and the minimum learning rate is set to 1e-5. In addition, all models adopt the Adam optimization algorithm with cross-entropy loss. All experiments are conducted with ten epochs, weight decay set to 1e-4, and momentum value set to 0.9. The backbone model is initialized with the weight parameters trained in ImageNet, and the network adopted in the experimental part is EFCOMFF-Net-v1 unless otherwise specified.

*2) Evaluation Metrics:* To evaluate that proposed model effectively, two widely used evaluation metrics are adopted, namely, OA and confusion matrix (CM). OA represents the ratio of the correctly predicted image to the overall image. All experiments in this work adopt $k$-fold cross-validation for training and testing, and the average of the five testing accuracies is adopted as the final experimental result. Furthermore, to analyze the detailed classification error between different classes, a CM is adopted to visualize it. Each row of the matrix represents all samples of the predicted class, and each column represents the samples of the ground-truth class.

*3) Hardware Platform and Software Environment:* All experiments are implemented on a workstation with 64-GB memory Intel i9-11900K at 3.50-GHz CPU and 24-GB NVIDIA GeForce RTX 3090 GPU.

### C. Comparisons of the State-of-the-Art Methods

This section presents two variants EFCOMFF-Net-v1 and EFCOMFF-Net-v2 of the network EFCOMFF-Net using ResNet50, ResNet152, and DenseNet121 as the backbone network for comparison with the latest methods. The introduction of these compared methods is listed in Table II.

For comparison purposes, the training rate of the four datasets in this work is set to be the same as the above methods. In addition, the visualization results of all confusion matrices in the experiment are obtained based on the ResNet152 backbone network adopting EFCOMFF-Net-v1 on the UCM dataset with 80% training rate, the AID dataset with 20% training rate, the NWPU dataset with 20% training rate, and the OPTIMAL-31 dataset with 80% training rate.

*1) Analysis of Experimental Results for the UCM Dataset:* The OA results of the experiments are shown in Table III. EFCOMFF-Net-v1 with ResNet152 as the backbone network achieves the best classification accuracies 99.35% and 99.85% on the UCM dataset with 50% and 80% training rates, respectively. Compared with the backbone network, the OA results are improved by 1.16% and 1.00%, respectively. Experiments show that more discriminative features can be obtained by integrating multiscale features that enhance information interaction and eliminating redundant information of deep features by using an attention mechanism. AFDP and ACGLNet adopt double branch networks as feature extractors, which will lead to overlapping features and insufficient fusion in the fusion stage. GRMA-Net and VGG_MS2AP achieve stable and excellent results by combining a single network with multiscale features and attention. However, the latter only sets different receptive fields at the end of CNN to obtain global features, ignoring the shallow local information, and the overall classification effect is slightly worse than the former. In addition, when ResNet50 and DenseNet121 are adopted as the backbone networks, more advanced classification accuracy is also achieved compared to the latest methods. As shown in Table III, because of the superiority of deep neural networks, the classification accuracy increases with deeper network depth. The depth feature approach has significantly improved the performance of scene classification for remote sensing images. The designed EFCOMFF-Net-v2 with ResNet50 and DenseNet121 as the backbone network has 0.06% and 0.14% improvement compared to EFCOMFF-Net-v1 at a 50% training rate, respectively, which indicates reducing the multiscale feature gap could effectively improve the classification performance.

The classification accuracy of each category in the UCM dataset by CM is visualized in Fig. 6. Due to the similar spatial layout and objects of buildings for "denseresidential" and "intersection," the visualization results show that they have 0.05% probability of being incorrectly predicted as "buildings." It can also be seen from Fig. 6 that the remaining other categories are identified 100% correctly.

*2) Analysis of Experimental Results for the AID Dataset:* The OA of the experiment is shown in Table IV. Among the methods combining different backbone networks proposed in this article, EFCOMFF-Net-v1 with ResNet152 as the backbone network achieves the best classification results. On the AID dataset with 20% and 50% training rates, the

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TABLE II
INTRODUCTION OF THE COMPARISON METHODS

| Methods | Backbone | Introduction |
|---|---|---|
| aggregating features from dual paths (AFDP) [45] | pre-trained MobileNetv2 and MobileNetv3 | Features are extracted by a double-branch network, and then the double-branch network is fused by the method of merging the bilinear pool and feature connection. |
| attention consistent network (ACNet) [46] | pre-trained VGG16 | The attention consistency model is designed to reduce intra-group differences and enhance inter-group differences. |
| lightweight channel attention and multiscale feature fusion discrimination (LmNet) [47] | pre-trained ResNeXt50 | Embedding channel attention mechanism into CNN to learn the salient features of important channels. |
| multi-scale staking attention pooling (VGG_MS2AP) [48] | pre-trained VGG-16 | The attention model is adopted to mine key local semantics in the feature graph and an extended convolution operator is adopted to extract multi-scale features. |
| multiscale dense networks with squeeze and excitation attention (SEMSDNet) [49] | pre-trained DenseNet121 | Automatic conversion between small network and deep network according to the complexity of test samples and the limitation of computing resources. |
| gated recurrent multi-attention network (GRMA-Net) [50] | pre-trained ResNet101 | Multi-scale features extracted by a multi-level attention module are sent to a deep-gated recurrent unit (GRU) [57] to capture long-range correlation and contextual relationships. |
| attention cascade global-Local network (ACGLNet) [51] | pre-trained ResNet50 and DenseNet121 | The double-branch structure is used as feature extractor and the local attention features and high-level features are fused and classified by FPN. |
| complete frequency channel attention network (CFCANet) [52] | pre-trained ResNet50 | The acquired low-frequency information of the image fully interacts with the feature map, which improves the processing ability of the network for remote sensing images with noise. |
| embedded self-distillation in compact multibranch ensemble network (ESD-MBENet) [53] | pre-trained ResNet50 | The weight sharing potential of multi-branch network is explored, and feature enhancement module is added to make up for the lack of diversity. |
| high-order graph convolutional network (H-GCN) [54] | pre-trained DenseNet121 | A high-order graph convolution network is proposed to study class dependency, and the attention mechanism is used to pay attention to the key components in the image, and the two features are fused to obtain distinctive feature representation. |
| remote sensing transformer (TRS) [55] | ResNet50 | The proposed self-attention is integrated into ResNet, and then several pure transformer encoders are connected in sequence to improve the model's ability to learn the overall features of the image. |
| enhanced attention module (EAM) [56] | pre-trained ResNet101 | Using the enhanced attention module to make the network learn more discriminant features. |
| V16_21k [19] | pre-trained Transformer | This paper explores the multi-head attention mechanism as the main building block to derive the long-term context relationship between pixels in an image. |

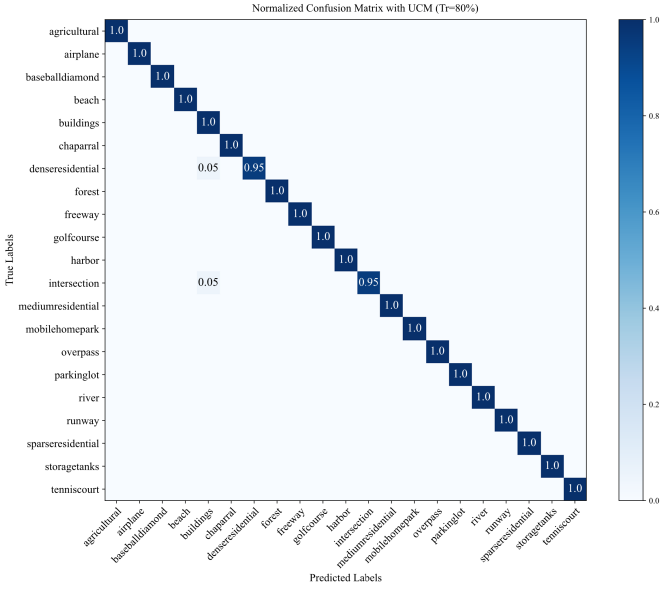


Fig. 6. CM of UCM dataset at 80% training rate.

classification accuracy is 96.44% and 98.70%, respectively, which is 1.88% and 2.1% higher than the backbone network. The better results also confirm that the deep network is effective in feature extraction. Compared with the latest method, AFDP achieves the best classification accuracy of 96.78% at the 20% training rate, and the proposed method achieves the second classification accuracy. But at 50% training rate, our method achieves the best classification accuracy, which is 0.84% higher than AFDP, which proves the stability and superiority of our method. AFDP, ACNet, and ACGLNet all adopt dual-branch networks, and the problems of overlapping features and insufficient fusion in the fusion stage lead to poor classification results. ESD-MBENet has achieved OA similar to our method by integrating the middle layer and the final output, but for the final output of the backbone network, we propose that the FAAM attention module focuses on the key areas in the scene image content, which enhances the distinguishing ability of features. Similarly, in the AID dataset, the classification effects of EFCOMFF-Net-v1 and EFCOMFF-Net-v2 using different backbone networks are also explored. EFCOMFF-Net-v1 achieves the best classification accuracy in combining different backbone networks, which shows that it is more robust than EFCOMFF-Net-v2.

In addition, to further compare the differences between EFCOMFF-Net-v1 and EFCOMFF-Net-v2 in the classification of various categories, the CMs of them are visualized as shown in Figs. 7 and 8. It can be seen from the results that the classification accuracy of EFCOMFF-Net-v1 reaches 100% in five categories including “BareLand,” “beach,” “Forest,” “Mountain,” and “Stadium.” The classification accuracy of EFCOMFF-Net-v2 for categories “BaseballField,” “Forest,” and “Mountain” reaches 100%, which shows that these two methods are targeted for different categories, and the OA effect of EFCOMFF-Net-v1 is better than that of EFCOMFF-Net-v2.

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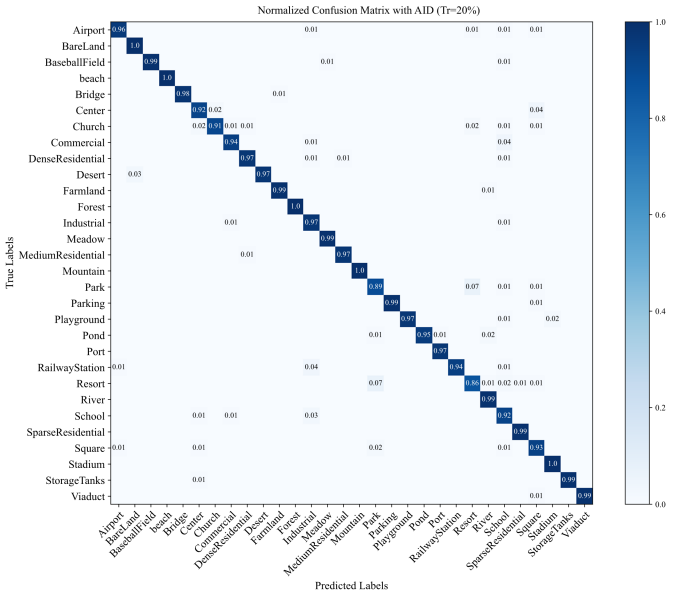


Fig. 7. EFCOMFF-Net-v1 uses AID dataset at 20% training rate of CM.

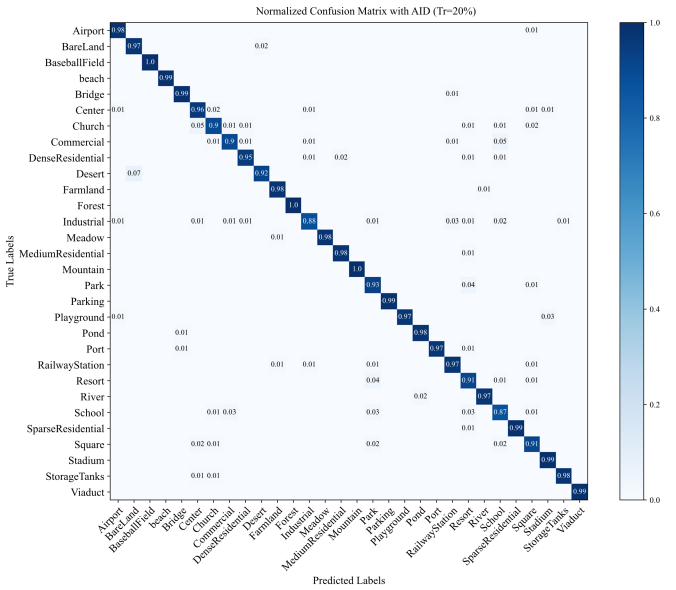


Fig. 8. EFCOMFF-Net-v2 uses AID dataset at 20% training rate of CM.

*3) Analysis of Experimental Results for the NWPU Dataset:* The OA of the experiments is shown in Table V. As always, among the proposed methods, EFCOMFF-Net-v1 with ResNet152 as the backbone network achieves the best classification accuracy of 93.62% and 95.88%, which is 1.76% and 1.68% higher compared to the backbone network, respectively. The best classification result is GRMA-Net with 93.67% OA at the 10% training rate compared with the current state-of-the-art methods. GRMA-Net also adopts a multilevel attention module to obtain multiscale feature information compared with our proposed networks. The difference is that it adopts the same attention structure to extract multiscale information and input it into the GRU [57] to capture long-range correlation and context relationship for classification, while we use targeted strategies to fuse multiscale features

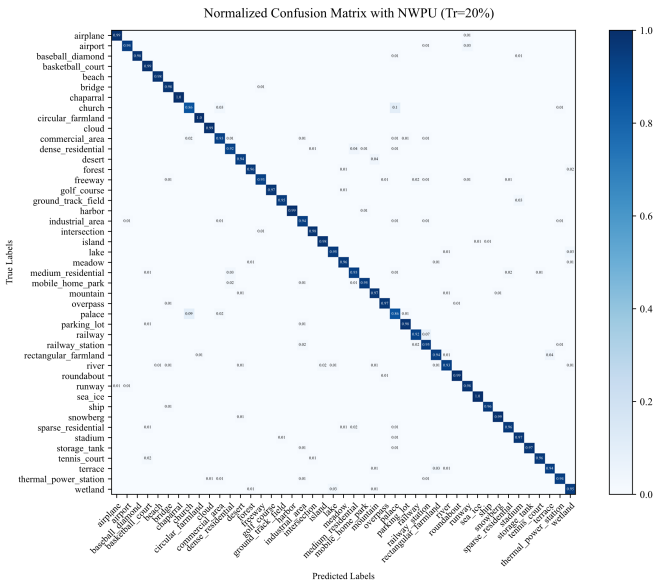


Fig. 9. CM of NWPU dataset at 20% training rate.

TABLE III

OA (%) OF DIFFERENT METHODS USING 50% AND 80% TRAINING RATES ON UCM DATASET, TR MEANS TRAINING RATE

| Methods | OA (%) | |
|---|---|---|
| | Tr = 50% | Tr = 80% |
| AFDP [45] | 98.81 ± 0.25 | 99.27 ± 0.45 |
| ACNet [46] | - | 99.76 ± 0.10 |
| LmNet [47] | 98.57 ± 0.38 | 99.52 ± 0.24 |
| VGG_MS2AP [48] | 99.09 ± 0.36 | 99.45 ± 0.32 |
| SEMSDNet [49] | 98.80 ± 0.59 | 99.41 ± 0.14 |
| GRMA-Net [50] | 99.29 ± 0.29 | 99.38 ± 0.18 |
| ACGLNet [51] | 98.14 ± 0.25 | 99.46 ± 0.12 |
| EFCOMFF-Net-v1 (ResNet50) | 99.23 ± 0.24 | 99.85 ± 0.18 |
| EFCOMFF-Net-v2 (ResNet50) | 99.29 ± 0.16 | 99.81 ± 0.10 |
| EFCOMFF-Net-v1 (ResNet152) | **99.35 ± 0.10** | **99.85 ± 0.18** |
| EFCOMFF-Net-v2 (ResNet152) | 99.32 ± 0.23 | 99.81 ± 0.14 |
| EFCOMFF-Net-v1 (DenseNet121) | 99.09 ± 0.21 | 99.80 ± 0.17 |
| EFCOMFF-Net-v2 (DenseNet121) | 99.23 ± 0.23 | 99.81 ± 0.13 |

and eliminate redundant information. The experimental results show that when the training rate increases, the advantages of our method are obvious, which indicates that with the increase of training samples, the semantic relevance is fully captured and the representation ability of features is strengthened. ESD-MBENet also adopts the strategy of extracting multiscale feature information. The experimental results in Table V show that the network adopting multiscale feature information for remote sensing image classification performs better in terms of accuracy and stability for OA at different training rates compared to other methods. Hence, extracting multiscale features is beneficial to significantly improve the classification accuracy.

As visualized in Fig. 9 for each category of the NWPU dataset, the classification accuracy of each category is above 85%, and the OA of 30 categories is above 95%. Only "Church" (86%) and "Palace" (86%) have OA lower than

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TABLE IV

OA (%) OF DIFFERENT METHODS USING 20% AND 50% TRAINING RATES ON AID DATASET

| Methods | OA (%) | |
|---|---|---|
| | Tr = 20% | Tr = 50% |
| AFDP [45] | **96.78 ± 0.13** | 97.86 ± 0.15 |
| ACNet [46] | 93.33 ± 0.29 | 95.38 ± 0.29 |
| CFCANet [52] | 93.76 ± 0.21 | 96.92 ± 0.16 |
| ESD-MBENet [53] | 96.00 ± 0.15 | 98.54 ± 0.17 |
| SEMSDNet [49] | 94.23 ± 0.63 | 97.64 ± 0.51 |
| GRMA-Net [50] | 96.19 ± 0.48 | 97.84 ± 0.39 |
| ACGLNet [51] | 94.44 ± 0.10 | 96.10 ± 0.10 |
| EFCOMFF-Net-v1 (ResNet50) | 96.28 ± 0.09 | 98.58 ± 0.10 |
| EFCOMFF-Net-v2 (ResNet50) | 96.11 ± 0.06 | 98.56 ± 0.08 |
| EFCOMFF-Net-v1 (ResNet152) | **96.44 ± 0.07** | **98.70 ± 0.07** |
| EFCOMFF-Net-v2 (ResNet152) | 96.38 ± 0.12 | 98.67 ± 0.10 |
| EFCOMFF-Net-v1 (DenseNet121) | 95.86 ± 0.13 | 98.19 ± 0.10 |
| EFCOMFF-Net-v2 (DenseNet121) | 95.69 ± 0.15 | 98.18 ± 0.13 |

TABLE V

OA (%) OF DIFFERENT METHODS USING 10% AND 20% TRAINING RATES ON NWPU DATASET

| Methods | OA (%) | |
|---|---|---|
| | Tr = 10% | Tr = 20% |
| AFDP [45] | 93.32 ± 0.11 | 95.07 ± 0.09 |
| ACNet [46] | 91.09 ± 0.13 | 92.42 ± 0.16 |
| SEMSDNet [49] | 91.68 ± 0.39 | 93.89 ± 0.63 |
| GRMA-Net [50] | **93.67 ± 0.21** | 95.32 ± 0.28 |
| CFCANet [52] | 91.72 ± 0.28 | 94.10 ± 0.20 |
| ESD-MBENet [53] | 92.50 ± 0.22 | 95.58 ± 0.08 |
| H-GCN [54] | 91.39 ± 0.19 | 93.62 ± 0.28 |
| TRS [55] | 93.06 ± 0.11 | 95.56 ± 0.20 |
| EFCOMFF-Net-v1 (ResNet50) | 93.17 ± 0.13 | 95.70 ± 0.12 |
| EFCOMFF-Net-v2 (ResNet50) | 93.03 ± 0.10 | 95.52 ± 0.13 |
| EFCOMFF-Net-v1 (ResNet152) | **93.62 ± 0.14** | **95.88 ± 0.11** |
| EFCOMFF-Net-v2 (ResNet152) | 93.50 ± 0.12 | 95.82 ± 0.12 |
| EFCOMFF-Net-v1 (DenseNet121) | 92.40 ± 0.15 | 95.19 ± 0.14 |
| EFCOMFF-Net-v2 (DenseNet121) | 92.36 ± 0.12 | 95.18 ± 0.12 |

90%, which is due to their similarity in architectural style and the wrong classification caused by the angle and size of aerial photography. In addition, in the most complex and largest remote sensing datasets, the classification OA of our method still reaches 100% in three categories: "chaparral," "circular_farmland," and "sea_ice." It can be known that strengthening the correlation of shallow features and introducing attention to deep features can improve the discrimination performance and realize the separability of different categories of complex images.

*4) Analysis of Experimental Results for the OPTIMAL-31 Dataset:* From the test results in Table VI, it is clear that the proposed EFCOMFF-Net-v1 with ResNet152 as the backbone network achieves the highest classification accuracy of 99.46%, which is 6.72% improvement compared to the backbone network and 2.04% improvement compared to the latest method GRMA-Net, which is a significant improvement for such challenging. In addition, other combinations of the proposed method also outperform the latest method by at least

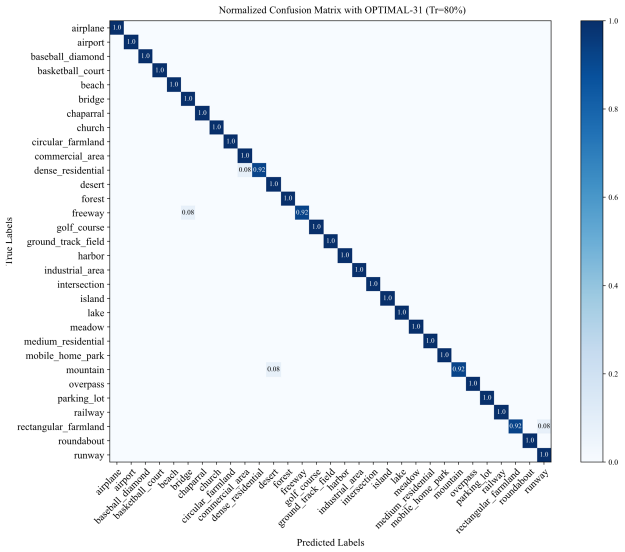


Fig. 10. CM of OPTIMAL-31 dataset at 80% training rate.

TABLE VI

OA (%) OF DIFFERENT METHODS USING 80% TRAINING RATE ON OPTIMAL-31 DATASET

| Methods | OA (%) |
|---|---|
| | Tr = 80% |
| GRMA-Net [50] | 97.42 ± 0.46 |
| ACGLNet [51] | 96.02 ± 0.24 |
| TRS [55] | 95.97 ± 0.13 |
| EAM [56] | 96.45 ± 0.28 |
| V16_21k [19] | 95.56 ± 0.18 |
| EFCOMFF-Net-v1 (ResNet50) | 99.35 ± 0.13 |
| EFCOMFF-Net-v2 (ResNet50) | 99.28 ± 0.16 |
| EFCOMFF-Net-v1 (ResNet152) | **99.46 ± 0.15** |
| EFCOMFF-Net-v2 (ResNet152) | 99.43 ± 0.13 |
| EFCOMFF-Net-v1 (DenseNet121) | 98.49 ± 0.16 |
| EFCOMFF-Net-v2 (DenseNet121) | 98.60 ± 0.14 |

TABLE VII

COMPARISON OF THE COMPLEXITY FOR EFCOMFF-NET-V1 AND EFCOMFF-NET-V2 ON DIFFERENT BACKBONE NETWORKS

| Methods | OA (%) | Params (M) | Flops (G) |
|---|---|---|---|
| EFCOMFF-Net-v1 (ResNet50) | 96.28 ± 0.09 | 44.79 | 8.73 |
| EFCOMFF-Net-v2 (ResNet50) | 96.11 ± 0.06 | 52.10 | 8.63 |
| EFCOMFF-Net-v1 (ResNet152) | 96.44 ± 0.07 | 77.43 | 23.63 |
| EFCOMFF-Net-v2 (ResNet152) | 96.38 ± 0.12 | 84.73 | 23.53 |
| EFCOMFF-Net-v1 (DenseNet121) | 95.86 ± 0.13 | 19.94 | 6.02 |
| EFCOMFF-Net-v2 (DenseNet121) | 95.69 ± 0.15 | 27.75 | 5.93 |

1.04%, once again demonstrating our method effectiveness. This is because the proposed EFCOMFF-Net enhances the information relevance of multiscale features and exploits the correlation between deep semantic information and shallow feature information to improve the abilities of discriminating salient features and improving classification effectiveness. Compared with V16_21k using transformer as the backbone

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TABLE VIII
TRAINING AND TESTING TIMES OF FILLING AND STRIDE ON UCM DATASET WITH 80% TRAINING RATE

| Backbone | Components | Split method | |
|---|---|---|---|
| | | Train (s) | Test (s) |
| ResNet50 | PAD1 | 15.75 | 2.47 |
| | PAD2 | 10.41 | 1.90 |
| | PAD3 | 9.67 | 1.86 |
| | PAD4 | 9.63 | 1.85 |
| | PAD5 | 9.50 | 1.85 |
| | PAD6 | 9.33 | 1.84 |

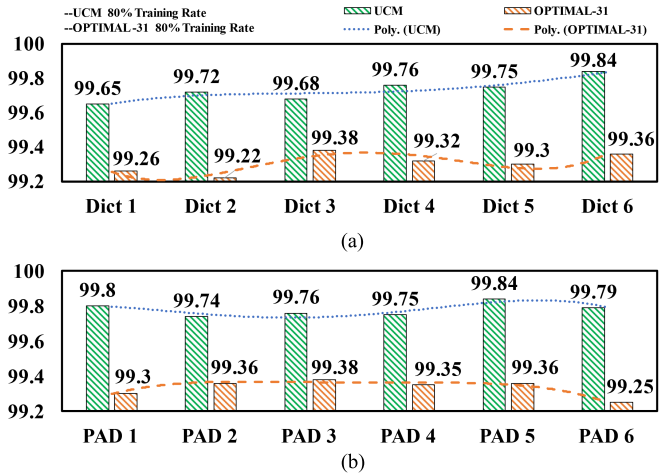


Fig. 11. Parameter discussion of FCEM: (a) expansion rate and (b) filling and stride length.

network, the methods based on CNN GRMA-Net, ACGLNet, and EAM have achieved better results. This is because the methods based on transformer pay more attention to the global features, while most of the remote sensing images are small in scale, so the local features are more important. In contrast, CNN can better handle the local features. The network TRS, which combines CNN and transformer, has achieved better classification performance than V16_21k, which shows that the fusion of local features and global features is beneficial to improve the classification performance.

Fig. 10 shows the visualization of the classification of each category; only the four categories "dense_residential" (92%), "freeway" (92%), "mountain" (92%), and "rectangular_farmland" (92%) are wrongly classified, which is attributed to the rationality of FAAM, FCEM, and FRM proposed by us.

### D. Comparison of EFCOMFF-Net-v1 and EFCOMFF-Net-v2

To further investigate the performance comparison between EFCOMFF-Net-v1 and EFCOMFF-Net-v2, we compare the number of model parameters and floating-point operations on the AID dataset with 20% training rate. As shown in Table VII, EFCOMFF-Net-v1 performs better than EFCOMFF-Net-v2 in terms of classification accuracy and number of parameters, but consumes more floating-point operations in inference. Overall, enhancing the relevance of deep features in scene classification

TABLE IX
TRAINING AND TESTING TIMES OF FILLING AND STRIDE ON THE OPTIMAL-31 DATASET WITH 80% TRAINING RATE

| Backbone | Components | Split method | |
|---|---|---|---|
| | | Train (s) | Test (s) |
| ResNet50 | PAD1 | 14.93 | 2.56 |
| | PAD2 | 10.01 | 1.96 |
| | PAD3 | 9.45 | 1.90 |
| | PAD4 | 9.14 | 1.82 |
| | PAD5 | 9.01 | 1.75 |
| | PAD6 | 8.95 | 1.64 |

not only leads to an increase in the number of parameters but also decreases the classification accuracy.

### E. Discussion on the Rationality of FCEM Parameters

The rationality of the design parameters of FCEM is discussed in the following. Considering that the shallow convolution block mainly extracts the local texture information of the image, and dilation rate becomes a key factor affecting the feature extraction, when dilation rate is too small, the receptive field becomes smaller, which is not conducive to obtaining more detailed information; the following kinds of dilation rates for all 3 × 3 convolutions in FCEM are discussed: Dict1: [1/1/1/1/1], Dict2: [2/2/2/2/2], Dict3: [3/3/3/3/3], Dict4: [3/3/5/5/3], Dict5: [2/2/5/7/2], and Dict6: [3/3/5/7/3]. As shown in Fig. 11(a), the experiment tests OA on the UCM and OPTIMAL-31 datasets. The parameter setting Dict6 gets the best performance on the UCM dataset with a training rate of 80%, and the Dict3 gets the best result on the OPTIMAL-31 dataset with an 80% training rate. After comprehensive consideration, the Dict6 with the highest average accuracy of the two datasets is adopted as the dilation rate of convolution for FCEM.

Similarly, for obtaining more feature information and reducing the model training time, several discussions on the stride and padding (stride and padding are kept the same) of all 3 × 3 convolutions in FCEM are carried out as follows: PAD1: [1/1/1/1/1], PAD2: [2/2/2/2/2], PAD3: [3/3/3/3/3], PAD4: [3/3/5/5/3], PAD5: [3/3/5/7/3], and PAD6: [3/3/7/7/3]. As shown in Fig. 11(b), the OA of the six parameter settings is also tested on the UCM and OPTIMAL-31 datasets with an 80% training rate, and PAD5 has the highest results. In addition, the training and testing times (the average of ten epochs) decrease with the increase of padding and stride, as in Tables VIII and IX. The parameters of PAD5 are selected as the padding and stride of FCEM, considering the training time and OA comprehensively.

### F. Discussion and Analysis of FCEM Structure

This section discusses different structural designs and multilevel feature combination approaches for FCEM. First, as shown in the FCEM module in Fig. 1, the rationality of the design of structures ①–③ is discussed. As shown in Table X, the designed model achieves the best OA on NWPU and

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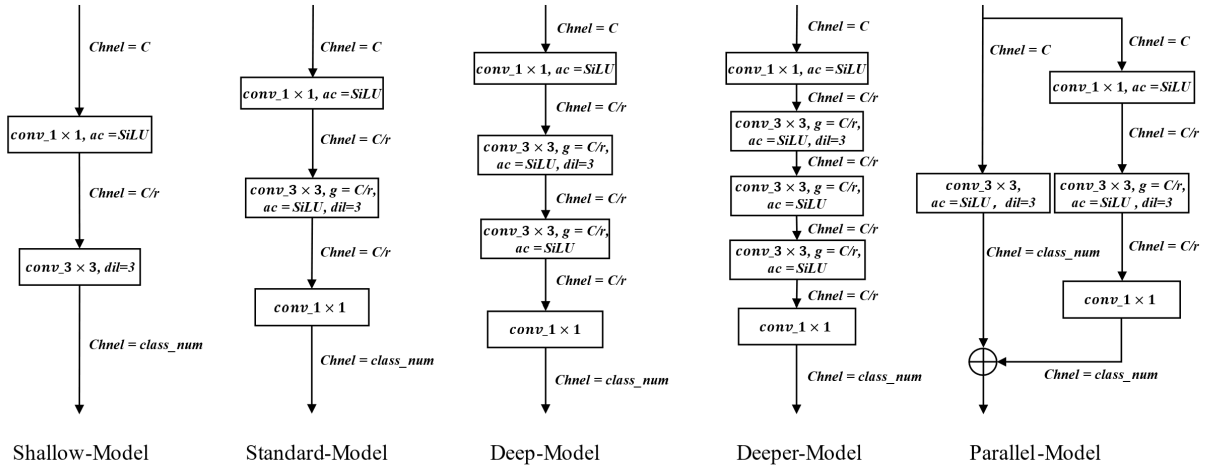


Fig. 12. Some feature fusion modules of different combinations.

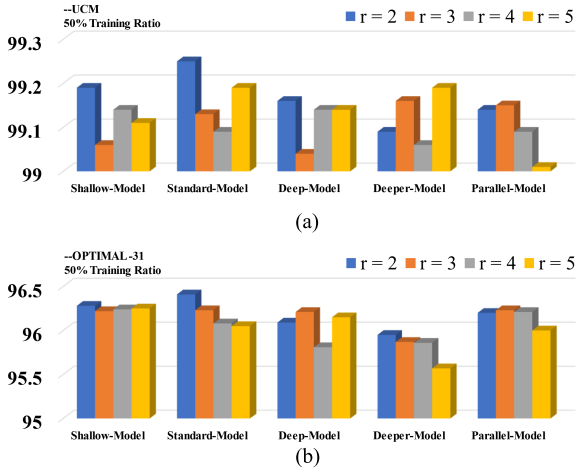


Fig. 13. OA of different channels of FRM on UCM and OPTIMAL-31 datasets (50% training rate): (a) OA for UCM dataset and (b) OA for OPTIMAL-31 dataset.

TABLE X

OA OF DIFFERENT FEATURE EXTRACTION STRUCTURES ON NWPU AND AID DATASETS (50% TRAINING RATE), W/O MEANS THERE IS NO SUCH STRUCTURE

| Backbone | Components | OA (%) | |
|---|---|---|---|
| | | NWPU (50%) | AID (50%) |
| ResNet50 | *w/o* ①, ③ | 97.57 ± 0.28 | 98.47 ± 0.24 |
| | *w/o* ② | 97.64 ± 0.16 | 98.38 ± 0.18 |
| | Ours | **97.89 ± 0.12** | **98.58 ± 0.10** |

AID datasets due to the difference in information relevance of multilevel features, and the structures ① and ② are designed to narrow the gap between shallow and deep-level features. On the one hand, the output of "Layer2" is upsampled and multiplied by the output of "Layer1." On the other hand, the output of "Layer1" is downsampled after passing through a convolution layer and multiplied by the output of "Layer2." In addition, while narrowing the gap and ensuring a small number of parameters, to avoid the great difference between shallow features and deep features cause the multiplication operation discards the key information of deep features, so the residual structure ③ is introduced.

We test the OA of different combinations of FCEM to extract multiscale information of ResNet50 and DenseNet121. Each structure of "Layer1"–"Layer3" is abbreviated as R1–R3, respectively. As shown in Table XI, when ResNet50 is the backbone feature extraction network without R1, the OA on NWPU and AID datasets is slightly inferior to the designed FCEM network in this work. The OA on NWPU dataset shows that the extracted feature information of R1 has little improvement on classification accuracy, while R2 and R3 play the major roles. The difference of OA for various combination methods of R1, R2, and R3 on the AID dataset is small, which shows that the proposed FCEM is unstable on extracting multilevel features of ResNet50. When DenseNet121 is adopted as the backbone feature extraction network, different combinations of R1, R2, and R3 play the same important role for the two datasets, which is caused by the differences of the structural designs. Comparing the convolution blocks of the two networks, DenseNet121 differs from ResNet50 in the design of residual blocks by adopting a concatenated approach to fuse features and a higher number of cycles. This differentiated design could avoid selective discarding of some layers, information blocking, etc., thus enhancing the adaptability to different datasets.

### G. Structure and Analysis of FRM

To investigate the rationality of designed FRM, Fig. 12 lists some convolutional blocks for different combinatorial approaches, where the increasing of network depth and the fusing features in a parallel manner are gradually discussed in this section. As shown in Table XII, the OA with different combinatorial approaches is tested on the UCM dataset with 80% and 50% training rates, respectively. The structural design of standard model achieves the best classification accuracy as shown by the experimental results. The distribution of results in Table XII shows that the classification accuracy does not increase linearly while the number of parameters increases as the network deepens.

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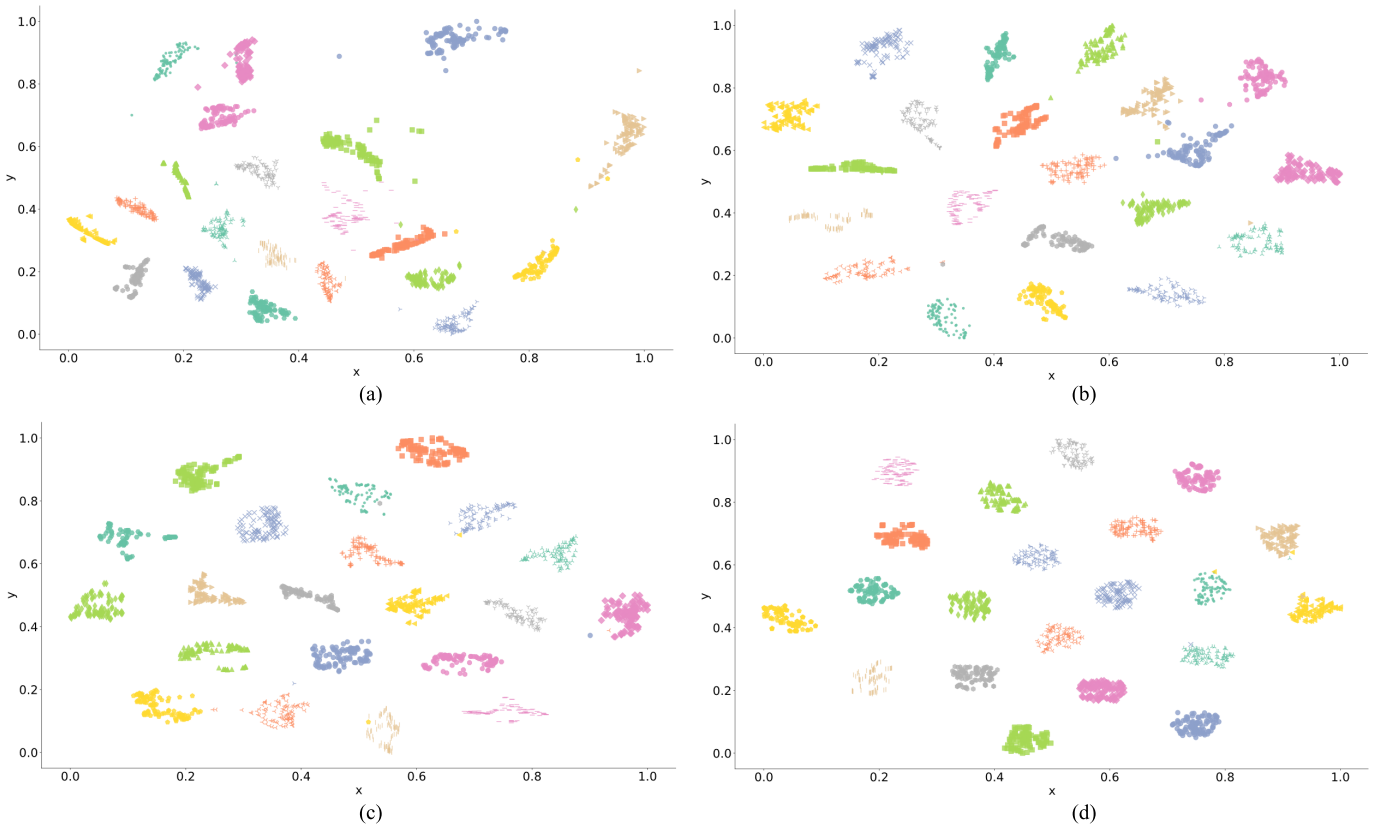


Fig. 14. Classification performance of different structure combinations using t-SNE on UCM dataset with an 80% training rate. Different colors and shapes represent different categories, and the same category is clustered into a small pile. The bigger the distance between small piles, the bigger the gap between different categories, and the better the classification effect: (a) backbone-ResNet50; (b) backbone-ResNet50 + FAAM; (c) backbone-ResNet50 + FAAM + FCEM; and (d) ours.

To further investigate the superiority of the designed FRM module, we explore the effect of the channel reduction multiplier $r$ on the classification accuracy of convolutional blocks with different combination methods. As shown in Fig. 13, the cases of that $r$ equal to 2, 3, 4, and 5 are explored on the UCM and OPTIMAL-31 datasets with a 50% training rate, respectively. The experimental results indicate that the best classification accuracy is achieved on both datasets when $r$ is equal to 2. Among different combined convolutional blocks, the standard model achieves the best classification results with the lower number of parameters. Therefore, we adopt it as the feature fusion module FRM.

### H. Ablation Experiments

To study the robustness of different designed feature extraction modules, the following ablation experiments are conducted.

*1) OA Comparison of Different Combined Feature Extraction Modules:* To demonstrate the generalization of the designed feature extraction module, ablation experiments were conducted on the three backbone networks ResNet50, DenseNet121, and ResNet152 on the NWPU and AID datasets with 20% training rate, respectively. The results are shown in Table XIII, which indicate that the OA of the designed network on the NWPU dataset is improved by 1.94%, 1.7%, and 1.68%, respectively, compared to the three backbone networks.

TABLE XI

USING RESNET50 AND DENSENET121 TO CALCULATE OA FOR DIFFERENT COMBINATIONS OF MULTILEVEL FEATURES ON NWPU AND AID DATASETS (50% TRAINING RATE)

| Backbone | Components | OA (%) | |
|---|---|---|---|
| | | NWPU (50%) | AID (50%) |
| ResNet50 | *w/o* R1 | 97.78 ± 0.24 | 98.38 ± 0.26 |
| | *w/o* R2 | 95.74 ± 0.18 | 98.32 ± 0.20 |
| | *w/o* R3 | 95.90 ± 0.23 | 98.44 ± 0.22 |
| | Ours | **97.88 ± 0.20** | **98.58 ± 0.10** |
| DenseNet121 | *w/o* R1 | 97.65 ± 0.24 | 97.97 ± 0.18 |
| | *w/o* R2 | 97.53 ± 0.18 | 97.87 ± 0.22 |
| | *w/o* R3 | 97.56 ± 0.25 | 97.94 ± 0.14 |
| | Ours | **97.84 ± 0.13** | **98.19 ± 0.10** |

The OA on the AID dataset is improved by 1.83%, 1.65%, and 1.88%, respectively, compared to the backbone networks. These statistics prove that the three designed feature extraction modules effectively extract the salient feature information. In addition, the experimental results show that the FCEM and FRM modules play the most critical role in the three backbone networks, which proves the effectiveness of our proposed multiscale integration strategy.

*2) Visual Analysis of EFCOMFF-Net and Backbone Network Using Different Feature Extraction Modules:* Using t-distributed stochastic neighbor embedding (t-SNE)

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TABLE XII

OA OF FEATURE FUSION MODULES FOR DIFFERENT COMBINATIONS ON UCM DATASETS WITH 80% AND 50% TRAINING RATES

| Backbone | Architecture | OA (%) | |
|---|---|---|---|
| | | UCM (80%) | UCM (50%) |
| ResNet50 | Shallow-Model | 99.83 ± 0.24 | 99.19 ± 0.21 |
| | **Standard-Model (ours)** | **99.85 ± 0.18** | **99.23 ± 0.24** |
| | Deep-Model | 99.81 ± 0.32 | 99.16 ± 0.26 |
| | Deeper-Model | 99.80 ± 0.14 | 99.09 ± 0.19 |
| | Parallel-Model | 99.80 ± 0.26 | 99.14 ± 0.23 |

TABLE XIII

OA OF DIFFERENT ARCHITECTURES IN THREE BACKBONE FEATURE EXTRACTION NETWORKS ON NWPU AND AID DATASETS (20% TRAINING RATES)

| Backbone | Components | OA (%) | |
|---|---|---|---|
| | | NWPU (20%) | AID (20%) |
| ResNet50 | Backbone | 93.76 ± 0.12 | 94.45 ± 0.13 |
| | FAAM | 94.10 ± 0.15 | 94.87 ± 0.15 |
| | FAAM + FCEM | 94.86 ± 0.17 | 95.73 ± 0.11 |
| | Ours | **95.70 ± 0.12** | **96.28 ± 0.09** |
| DenseNet121 | Backbone | 93.49 ± 0.14 | 94.21 ± 0.12 |
| | FAAM | 93.76 ± 0.18 | 94.52 ± 0.15 |
| | FAAM + FCEM | 94.34 ± 0.16 | 95.21 ± 0.18 |
| | Ours | **95.19 ± 0.14** | **95.86 ± 0.13** |
| ResNet152 | Backbone | 94.20 ± 0.20 | 94.56 ± 0.16 |
| | FAAM | 94.67 ± 0.17 | 94.89 ± 0.20 |
| | FAAM + FCEM | 95.26 ± 0.15 | 95.86 ± 0.12 |
| | Ours | **95.88 ± 0.11** | **96.44 ± 0.07** |

TABLE XIV

COMPLEXITY COMPARISON OF DIFFERENT COMBINATION FEATURE EXTRACTION MODULES ON THE AID DATASET WITH 20% TRAINING RATE

| Architecture | Params (M) | Train (s) | Test (min) | AID (20%) |
|---|---|---|---|---|
| *w/o* FAAM | 42.15 | 26.36 | 1.46 | 95.86 ± 0.14 |
| *w/o* FCEM | 30.85 | 22.87 | 1.21 | 95.42 ± 0.24 |
| *w/o* FRM | 41.77 | 25.98 | 1.44 | 95.73 ± 0.11 |
| EFCOMFF-Net (Ours) | 44.79 | 30.71 | 1.58 | **96.28 ± 0.09** |
| ResNet50 | 25.62 | 20.13 | 1.10 | 94.45 ± 0.13 |

technology, we map the network classification results in high-dimensional space to the 2-D characteristics for visualization. As shown in Fig. 14, we visualize the backbone network and add the results of different feature extraction blocks. Fig. 14(b) represents the visualization result of adding FAAM to the backbone network. Compared with Fig. 14(a), the classification error is reduced and the intraclass aggregation is higher, which means a better classification effect. Fig. 14(c) represents the network with the addition of FAAM and FCEM. Compared with the structure of Fig. 14(b), the classification error rate is reduced while the interclass distance is larger, and the larger interclass distance indicates a lower correlation of different classes, which is more favorable for classification. Fig. 14(d) shows the best results for the proposed EFCOMFF-Net in terms of classification accuracy, intraclass aggregation, and interclass distance compared to the backbone network.

*3) Complexity Analysis of Different Feature Extraction Modules:* The number of parameters and training time are usually used as indicators to measure the quality of the model. As shown in Table XIV, we test the parameter quantity, training time, testing time, and OA (the training time and testing time are the average of ten epochs) of different combination feature extraction modules based on the backbone network ResNet50. The results show that the total parameter number of EFCOMFF-Net-ResNet50 is 44.79M, which is 19.17M higher than the 25.62M of ResNet50. The number of parameters for FAAM, FCEM, and FRM is 2.64M, 13.94M, and 3.02M, respectively. The largest number of parameters is generated by FCEM, which is because FCEM has the multiple convolution blocks and involves the process of upsampling and increasing the number of channels, although the entire model structure is designed to minimize the number of parameters as much as possible. Moreover, with the improvement of model complexity, the training and testing times of the whole network inevitably increase, but they are all within a reasonable range.

## V. CONCLUSION

In this article, two network model structures, EFCOMFF-Net-v1 and EFCOMFF-Net-v2 are designed to explore whether narrowing the correlation between multiscale features could improve the performance of remote sensing image scene classification. EFCOMFF-Net-v1 focuses on establishing a correlation between shallow features, while EFCOMFF-Net-v2 is designed to explore whether deep feature correlation would improve the representation ability of the network.

From the analysis of experimental results, with the establishment of feature correlation from shallow layer to deep layer and the increase of parameters, except that the OA of EFCOMFF-Net-v2 is better than that of EFCOMFF-Net-v1 for the cases of ResNet50 and DenseNet121 as the backbone networks on the UCM with 50% training rate and OPTIMAL-31 with 80% training rate, the effect of EFCOMFF-Net-v1 in other combination methods is better than the later. This is because the gap between deep features and shallow features is too large, and forcing the gap between local contour information and deep semantic information would result in losing the detailed information of lower layers because of the difference in spatial distribution, thus affecting the classification effect. Furthermore, to tap the global information contained in deep features, FAAM is proposed to aggregate features and encode more accurate spatial information. Finally, FRM is developed to address the differences between deep and shallow features, and reduce the differences between them by recoding spatial information, thus smoothing the transition of fused features. The experimental results of the proposed EFCOMFF-Net on several datasets adopting three kinds of backbone networks verify the superiority and better robustness of our method.

We acknowledge that there are still some deficiencies in this study. As the proposed EFCOMFF-Net involves multiscale feature extraction and fusion which is inevitably accompanied

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by changes in the number of image scales and channels, the increase of parameters would result in slower training speed compared with some existing mainstream networks. In the future, we plan to improve the parameter configuration of convolution and reduce the scale change to optimize the network structure, reduce the network parameters, and increase the operation speed. The advantages of the proposed method in dealing with multiscale features can be easily applied to image segmentation and target detection.

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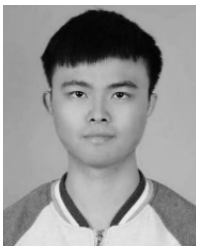

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